MAJOR AGRONOMIC PRACTICES FOR GROWING MATOOKE AND RABBITRY

(From selection of planting material and breeding stock up to harvest, including diseases, prevention and how to get high yields)

I have been involved in the school garden for the last two years, working on both matooke and rabbits. From what I have seen with my own eyes, what our agriculture teacher has taught us, and the few books and notes we use, I have put together this information. My aim is to write it in a clear and practical way so that any student or farmer who wants to start can follow without getting confused. I have also included some mistakes that we made at the beginning and how we corrected them.

GROWING MATOOKE

Matooke is the main food in many homes in Uganda, especially in the central and western regions. In school we grow it both for learning and for the kitchen. It is not planted from seeds like maize or beans. We use suckers or tissue culture plantlets.

1. Choosing good planting material

This is the most important step. If you start with diseased suckers, the whole garden will suffer later. Never take suckers from a garden that has banana wilt, weevil damage or yellowing leaves. Choose sword suckers – these are the ones with narrow pointed leaves and they are firmly attached to the mother plant. Water suckers have broad leaves and are weakly attached; they produce small bunches so we always remove them.

If possible, get tissue culture plantlets from a trusted nursery. They are clean and free from most diseases. In our school we started with ordinary suckers but later introduced a few improved types. The improved ones like some of the NARITA varieties give bigger bunches and resist black Sigatoka better. Local varieties such as Kibuzi or Mbwazirume also do well if the management is good.

Before planting, we clean the suckers by removing soil and cutting off any damaged roots or outer leaf sheaths. Some people dip them in hot water for a short time to kill nematodes, but we have not tried that much.

2. Land preparation and planting

Choose a place with deep soil that does not get waterlogged. Matooke does not like water sitting around the roots for long. Dig holes that are about 60 cm deep and 60 cm wide (roughly two feet by two feet). On sloping land we dig along the contour to reduce soil erosion.

Mix the top soil with plenty of well-rotted manure. In our garden we use a mixture of cow dung and goat droppings. About half a basin to one full basin per hole is good. If the soil is very poor you can add a little more. Plant at the beginning of the rainy season so that the plants get enough water to establish. Spacing of 3 metres by 3 metres works well for most varieties. This gives the plants enough space for the leaves and also makes it easy to move between them when weeding or harvesting.

After planting we sometimes intercrop with beans or groundnuts in the first few months. The beans help to cover the soil and also add some nitrogen. Once the banana leaves start to shade the ground heavily we stop the intercropping.

3. Management practices that increase yield

Desuckering is very important. On one mat we leave only three plants – the mother (the one carrying or about to carry a bunch), the daughter and the granddaughter. If you leave more than three, the plants compete for food and water and the bunches become small. We remove the extra suckers every two to three weeks, especially during the rainy season when they grow fast.

Mulching helps a lot. We use dry grass, maize stalks or sometimes banana leaves that have been dried. The mulch keeps the soil moist, reduces weeds and adds organic matter as it breaks down. Important point: do not pile the mulch right against the stem of the plant. Leave a small space around the base. If you cover the base completely, weevils like to hide there and the stem can also rot.

We apply manure at least once a year, preferably at the beginning of the rains. About 8 to 10 kg of well-rotted manure per plant is what we aim for. Bananas need a lot of potassium, so if the leaves start showing yellowing between the veins it may be a sign of shortage. In school we mainly rely on manure because chemical fertilisers are expensive.

Remove dry and dead leaves regularly but do not remove green leaves unless they are badly diseased. The green leaves are the ones that make food for the bunch. When the bunch has formed all its hands, we cut off the male bud (the flower at the end). This reduces the chance of insects carrying bacterial wilt into the plant.

Heavy bunches can break the stem, especially when there is strong wind. We use forked sticks to prop them up. It is simple but it saves many bunches.

One thing we learnt the hard way is tool hygiene. When you cut one plant and move to another without cleaning the knife or panga, you can easily spread banana wilt. Now we clean the tools with diluted bleach or by passing them through fire after every few plants.

4. Harvesting

A bunch is ready for harvest roughly four to five months after the flower appears. The exact time depends on the variety and the weather. The fingers should be full and the sharp edges should have become more rounded. We harvest when the bunch is still green because that is how matooke is preferred. Cut the bunch carefully and also cut the pseudostem so that it does not fall on other plants. After harvesting we chop the remaining stem into pieces and leave them to dry; later we can use them as mulch or just let them rot and return nutrients to the soil.

5. Common diseases, pests and how we control them

Banana bacterial wilt (also called BXW) is the most serious disease. The leaves turn yellow and wilt starting from the top, and when you cut the stem you see yellowish bacterial ooze. There is no chemical cure. The only way is to remove the infected plant completely, including the corm, and destroy it. Always remove the male bud early and sterilise tools. We also avoid moving suckers from infected gardens into clean ones.

Black Sigatoka appears as small dark spots on the leaves that later join together and kill large parts of the leaf. It reduces the ability of the plant to make food. Removing the badly affected leaves and planting resistant varieties helps. Good spacing also allows air to move freely and reduces the disease.

Banana weevils are a big problem in older gardens. The adult weevil lays eggs at the base of the plant and the larvae tunnel inside the corm. You may see sawdust-like material and the plant becomes weak and can topple. Control starts with clean planting material. We also remove old stumps after harvest and sometimes use split pseudostem traps to catch the adult weevils. Keeping the garden clean reduces their numbers.

Nematodes are tiny worms that attack the roots. The plants become stunted and can fall over easily. Using clean suckers and applying plenty of manure helps the plants to grow stronger roots.

Other problems include toppling due to weak roots or heavy bunches, and sometimes drought stress. Mulching and good soil fertility reduce these problems.

If management is good, a mat can continue producing for many years, although the bunches may become smaller after five or more years. In school we replace some of the oldest mats gradually.

RABBITRY

Keeping rabbits in the school garden has been interesting. They do not need a lot of space, they grow fast, and their manure is excellent for the banana garden. We started with a few rabbits and slowly increased the number.

1. Selecting breeding stock

Choose rabbits that look healthy. The eyes should be clear, the nose dry, the fur smooth, and the legs strong. Avoid rabbits that are sneezing or have discharge from the nose or eyes. For meat production, New Zealand White and Californian are popular because they grow fast and have good body size. Local crosses can also do well if they are selected carefully.

One mature buck can serve about 10 to 12 does. Do not keep too many bucks because they fight. When buying new rabbits, keep them separate for at least two weeks to make sure they are not carrying disease into the main group.

2. Housing

We built raised hutches using timber and wire mesh. The floor is wire so that the droppings fall down and the rabbits stay clean and dry. Under the hutches we collect the manure for the garden. The hutches must protect the rabbits from rain, direct strong sun, strong wind, and animals like dogs or snakes. Each breeding doe needs enough space and a nest box when she is about to give birth. The nest box is filled with dry grass or soft material so that the kits are warm.

Good ventilation is very important. If the air is stale and full of ammonia from urine, the rabbits easily get respiratory problems.

3. Breeding and management of young ones

The doe is taken to the buck's cage for mating, not the other way round. Mating usually takes a short time. Write down the date so that you know when the kits are expected. Pregnancy lasts about 30 to 31 days. About three to five days before kindling, put the nest box in the doe's hutch.

A good doe can produce between 6 and 10 kits, sometimes more. The kits are born without fur and with closed eyes. The mother looks after them. We check quietly to make sure all are feeding. After about five to six weeks we wean them by removing them from the mother. If the doe is in good condition she can be re-mated soon after weaning. In our experience, aiming for four to five litters per year is realistic without stressing the doe too much.

The young rabbits grow fast if feeding is good. We separate the males and females when they are about two to three months to avoid unwanted mating.

4. Feeding

Rabbits need plenty of fibre. We give them grass, sweet potato vines, cabbage leaves, and other green materials that are not poisonous. We also give commercial pellets when available, especially to the growing ones and lactating does. Change of feed should be done gradually, otherwise they can get diarrhoea. Clean water must be available at all times. We use simple drinkers that do not tip over easily.

Overfeeding adults with concentrates can make them fat and reduce breeding performance, so we control the amount for the breeding stock.

5. Diseases and how we prevent them

The most common problem we have faced is snuffles. The rabbit sneezes and has discharge from the nose. It spreads easily in poorly ventilated houses. Keeping the hutches clean and well ventilated is the best prevention. Sometimes the teacher helps us with treatment, but prevention is better.

Coccidiosis causes diarrhoea, especially in young rabbits. It is linked to dirty and wet conditions. Cleaning the hutches regularly and keeping the floor dry reduces it a lot.

Ear mites make the rabbit shake its head and scratch the ears. The ears may have crusts. We treat it early when we notice it.

Sore hocks (wounds on the feet) happen when the wire floor is too rough or the rabbit is too heavy. Providing a small wooden board for resting helps.

Other problems include diarrhoea from sudden change of feed and sometimes wounds from fighting. We isolate any sick rabbit immediately so that the problem does not spread.

6. Harvesting (slaughter)

Rabbits are usually ready for slaughter when they reach about 2 to 2.5 kg live weight. This can take 10 to 14 weeks depending on feeding and breed. The meat is white, soft and has less fat compared to other meats. After slaughter the skin can be dried if someone wants to use it. The manure is collected, composted for some time, and then applied to the matooke garden. We have noticed that bananas that receive rabbit manure grow very well.

INTEGRATION OF THE TWO ENTERPRISES IN THE SCHOOL GARDEN

One of the best things is that the two projects support each other. Rabbit manure is rich and improves the soil for matooke. In return, some of the banana leaves and pseudostems can be used as feed or bedding after proper preparation. Students learn both crop and animal husbandry at the same time. We keep simple records of planting dates, desuckering, mating dates, litter sizes and harvest weights. These records help us to see what is working and what needs improvement.

COMMON MISTAKES WE MADE AND LESSONS LEARNT

At the beginning we left too many suckers on the banana mats and the bunches were small. We also used to pile mulch against the stems and later found more weevil damage. With rabbits, we once put new rabbits straight into the main hutches without quarantine and introduced a respiratory problem. Cleaning tools between banana plants was something we ignored until we saw wilt spreading. All these taught us that small details matter a lot.

CONCLUSION

Growing matooke and keeping rabbits can give good results if the basic practices are followed carefully. The most important points are: start with clean and healthy planting material or breeding stock, keep the garden and hutches clean, manage the number of plants per mat and the breeding of rabbits properly, and apply manure regularly. Diseases are easier to prevent than to cure. In a school setting these projects also teach responsibility, record keeping and the link between crops and animals.

I hope this information will help other students and farmers who want to start. What I have written is based on what we do in our school garden and what has worked for us. Different places may have slight differences in soil and weather, so it is always good to observe and adjust.